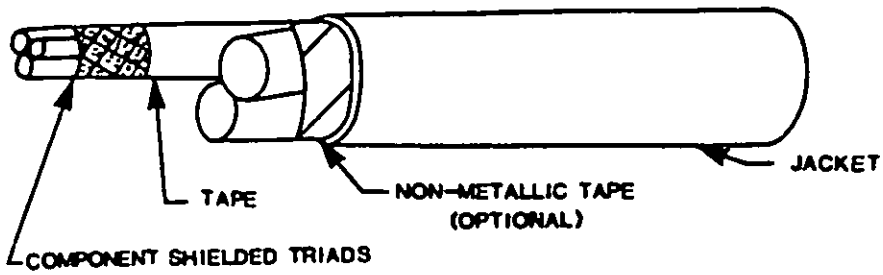
MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-TRIAD, AWG 18, TYPE 3XS (INCLUDING VARIATION TYPE 3XSA)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24640.



SH 12758

COMPONENT SHIELDED TRIAD: One black MIL-W-81044/12-18 wire and one white MIL-W-81044/12-18 wire twisted with one red MIL-W-81044/12-18 wire, except identification codes and methods shall be in accordance with MIL-C-24640. The shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (non-watertight).

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TABLE I. Construction details - unarmored.

Type number	Number of triads	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
3XS-7	7	0.044	0.601	0.647	294.0

TABLE II. Construction details - armored.

Type number	Number of triads	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
			(min)	(max)	
3XSA-7	7	0.044	0.651	0.697	346.9

REQUIREMENTS:

Qualification required.
Construction details: See tables I and II.

When this cable is manufactured with armor, the following additional requirements shall apply:

(a) Braided metal armor and paint shall be added.
(b) Cable surface marking will not be required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	6.35
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Conductor to shield	1000
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity	Not applicable
Jacket flaws	Not applicable

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	55
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	40 ± 5
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type 3XSA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance	
Milliohms per meter, maximum	Not applicable
EMP response, dB, minimum	Not applicable
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0903-11)

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